FOR A NODAL CURVE OF GENUS AT LEAST 2 WITH AMPLE DUALIZING SHEAF, THE TRICANONICAL SHEAF IS VERY AMPLE AND HAS NO FIRST COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the global-generation step, the descent to the algebraic closure, the full inequality reduction including both exceptional cases (ampleness excluding degree zero; Riemann-Roch handling Y = X), and the concluding vanishing argument are all retained verbatim.

Editorial note. The label lemma-tricanonical names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- No rational tails or bridges under ampleness. Varieties, Lemma varieties-lemma-ampleness-in-terms-of-degrees-components; Same chapter, Lemmas lemma-rational-tail-negative and lemma-rational-bridge-zero. Shows X contains no rational tails or bridges, which also rules out degree-zero restrictions of omega_X to reduced connected subschemes.
- Global generation of the tricanonical sheaf. Same chapter, Lemma lemma-contracting-rational-tails. Gives that omega_X^tensor 3 is globally generated once tails and bridges are absent.
- Separation criterion and descent to the algebraic closure. Varieties, Lemma varieties-lemma-variant-separate-points-tangent-vectors; Descent, Lemma descent-lemma-descending-property-closed-immersion. Reduces closed immerseness to surjectivity of H^0(X, L) -> H^0(Z, L|_Z) for every degree-2 subscheme Z, and lets one replace k by its algebraic closure.
- Vanishing on Gorenstein curves and the tail inequality. Same chapter, Lemmas lemma-vanishing-on-gorenstein and lemma-no-rational-tail; Duality for Schemes, Section duality-section-dualizing (Gorenstein hypothesis). Turns the surjectivity problem into the inequality 3 deg(omega_X|_Y) > -2 chi(Y, O_Y) + deg(Z cap Y) and supplies the bound away from the exceptional cases.

Source reference. Original section 01ND: sources/stacks-chow/tags.txt, source lines 2154–2154. Prerequisite pointer invoked at proof line(s) 6811; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02L6: sources/stacks-chow/tags.txt, source lines 3397–3397. Prerequisite pointer invoked at proof line(s) 6814; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3L: sources/stacks-curves/pinned-original-source.tex, source lines 6135–6135. Prerequisite pointer invoked at proof line(s) 6804; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3J: sources/stacks-curves/pinned-original-source.tex, source lines 6032–6032. Prerequisite pointer invoked at proof line(s) 6844; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E64: sources/stacks-curves/pinned-original-source.tex, source lines 6317–6317. Prerequisite pointer invoked at proof line(s) 6801; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E63: sources/stacks-curves/pinned-original-source.tex, source lines 5987–5987. Prerequisite pointer invoked at proof line(s) 6801; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS6: sources/stacks-curves/pinned-original-source.tex, source lines 1218–1218. Prerequisite pointer invoked at proof line(s) 6848; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3E: sources/stacks-curves/pinned-original-source.tex, source lines 5680–5680. Prerequisite pointer invoked at proof line(s) 6830; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B5Y: sources/stacks-chow/tags.txt, source lines 14051–14051. Prerequisite pointer invoked at proof line(s) 6800; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Prerequisite pointer invoked at proof line(s) 6838; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0E8T: sources/stacks-chow/tags.txt, source lines 17828–17828. Prerequisite pointer invoked at proof line(s) 6820; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k of dimension 1 with $H^0(X, \mathcal{O}_X) = k$ having genus $g \geq 2$. Assume the singularities of X are at-worst-nodal and ω_X is ample. Then $\omega_X^{\otimes 3}$ is very ample and $H^1(X, \omega_X^{\otimes 3}) = 0$.*

Proof. Combining Varieties, Lemma 0B5Y and Lemmas 0E63 and 0E64 we see that X contains no rational tails or bridges. Then we see that $\omega_X^{\otimes 3}$ is globally generated by Lemma 0E3L. Choose a k-basis $s_0, \ldots, s_n$ of $H^0(X, \omega_X^{\otimes 3})$. We get a morphism

$$\varphi_{\omega_X^{\otimes 3}, (s_0, \ldots, s_n)} : X \longrightarrow \mathbf{P}_k^n$$

See Constructions, Section 01ND. The lemma asserts that this morphism is a closed immersion. To check this we may replace k by its algebraic closure, see Descent, Lemma 02L6. Thus we may assume k is algebraically closed.

Assume k is algebraically closed. We will use Varieties, Lemma 0E8T to prove the lemma. Let $Z \subset X$ be a closed subscheme of degree 2 over Z with ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$. We have to show that

$$H^0(X, \mathcal{L}) \to H^0(Z, \mathcal{L}|_Z)$$

is surjective. Thus it suffices to show that $H^1(X, \mathcal{I}\mathcal{L}) = 0$. To do this we will use Lemma 0E3E. Thus it suffices to show that

$$3 \deg(\omega_X|_Y) > -2\chi(Y, \mathcal{O}_Y) + \deg(Z \cap Y)$$

for every reduced connected closed subscheme $Y \subset X$. Since k is algebraically closed and Y connected and reduced we have $H^0(Y, \mathcal{O}_Y) = k$ (Varieties, Lemma 0BUG). Hence $\chi(Y, \mathcal{O}_Y) = 1 - \dim H^1(Y, \mathcal{O}_Y)$. Thus we have to show

$$3 \deg(\omega_X|_Y) > -2 + 2 \dim H^1(Y, \mathcal{O}_Y) + \deg(Z \cap Y)$$

which is true by Lemma 0E3J except possibly if $Y = X$ or if $\deg(\omega_X|_Y) = 0$. Since ω_X is ample the second possibility does not occur (see first lemma cited in this proof). Finally, if $Y = X$ we can use Riemann-Roch (Lemma 0BS6) and the fact that $g \geq 2$ to see that the inquality holds. The same argument with $Z = \emptyset$ shows that $H^1(X, \omega_X^{\otimes 3}) = 0$. □